

WEEK 1 STATUS REPORT: TOPOLOGY ARCHITECTURE & IP ADDRESS MATRIX DESIGN

1. Project Scope & Architecture Mapping

- 1 Finalized the architectural design layout to resolve Problem 9 (Network Configuration Backup Automation).
- 2 Mapped out a 3-node core network router topology consisting of an active target matrix: R1, R2, and R3.
- 3 Provisioned a centralized multilayer management switch deployment to act as the primary connection backplane.
- 4 Identified and isolated a dedicated endpoint station to serve as the secure automation host executing the Python script framework.

2. Logical Subnet & IP Plan Allocation

- 1 Engineered a logical class-based network schema to guarantee structured, non-overlapping IP configurations across the infrastructure.
- 2 Defined the core management subnet boundary as **192.168.1.0/24**.
- 3 Formulated the exact physical and virtual node allocation matrix:
 - Router 1 (R1) Management Plane IP: **192.168.1.1 /24**
 - Router 2 (R2) Management Plane IP: **192.168.1.2 /24**
 - Router 3 (R3) Management Plane IP: **192.168.1.3 /24**
 - Automation Server Backup Node IP: **192.168.1.10 /24**
 - Baseline Management Subnet Mask: **255.255.255.0**
- 4 Initialized the tracking repository baseline on GitHub with localized target directories (**/configs, /scripts, /reports, /screenshots**).